

HP2 Analysis: FM action_weight=5

Four-way comparison — A: Diffusion S6 · B: FM S6 (aw=10) · C: FM S6 (aw=1) · D: FM S6 (aw=5, HP2)

A = Diffusion S6	B = FM aw=10 · original	C = FM aw=1 · HP1	D = FM aw=5 · HP2 (this file)
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D uses FM with **action_weight=5**, trained to checkpoint step 93000 (vs 89000 for C, same architecture). Primary question: does aw=5 split the difference between aw=1 (C) and aw=10 (B), or does it produce a distinct performance profile?

Figure 1 displays four heatmaps (A, B, C, D) showing GAC Success (0.0 to 1.0) for different models and datasets. The datasets are TR, TL, and Both. The models are grouped into four categories: Diffusion S6 (A), FM aw=10 (B), FM aw=1 (C), and FM aw=5 (D). The color scale indicates GAC Success, ranging from 0.0 (dark red) to 1.0 (dark green).

A: Diffusion S6

Model	TR	TL	Both
DiffS6	1.0	1.0	0.0
DiffS6+1	1.0	1.0	1.0
DiffS6+2	1.0	1.0	1.0
DiffS6+3	1.0	1.0	1.0
DiffS6+4	1.0	1.0	1.0
DiffS6+5	1.0	1.0	1.0
DiffS6+6	1.0	1.0	1.0
DiffS6+7	1.0	1.0	1.0
DiffS6+8	1.0	1.0	1.0
DiffS6+9	1.0	1.0	1.0
DiffS6+10	1.0	1.0	1.0
DiffS6+11	1.0	1.0	1.0
DiffS6+12	1.0	1.0	1.0
DiffS6+13	1.0	1.0	1.0
DiffS6+14	1.0	1.0	1.0
DiffS6+15	1.0	1.0	1.0
DiffS6+16	1.0	1.0	1.0

B: FM aw=10

Model	TR	TL	Both
DiffS6	1.0	1.0	0.0
DiffS6+1	1.0	1.0	1.0
DiffS6+2	1.0	1.0	1.0
DiffS6+3	1.0	1.0	1.0
DiffS6+4	1.0	1.0	1.0
DiffS6+5	1.0	1.0	1.0
DiffS6+6	1.0	1.0	1.0
DiffS6+7	1.0	1.0	1.0
DiffS6+8	1.0	1.0	1.0
DiffS6+9	1.0	1.0	1.0
DiffS6+10	1.0	1.0	1.0
DiffS6+11	1.0	1.0	1.0
DiffS6+12	1.0	1.0	1.0
DiffS6+13	1.0	1.0	1.0
DiffS6+14	1.0	1.0	1.0
DiffS6+15	1.0	1.0	1.0
DiffS6+16	1.0	1.0	1.0

C: FM aw=1

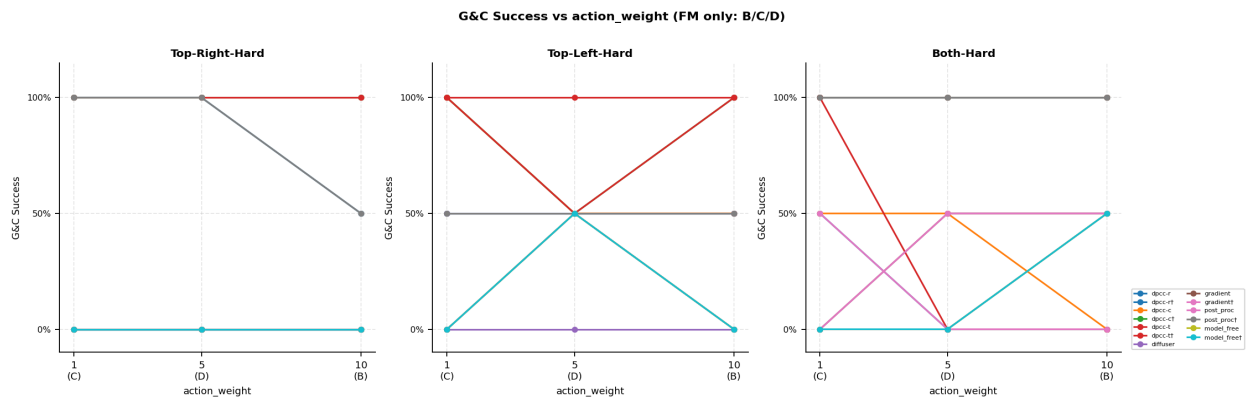
Model	TR	TL	Both
DiffS6	1.0	1.0	0.0
DiffS6+1	1.0	1.0	1.0
DiffS6+2	1.0	1.0	1.0
DiffS6+3	1.0	1.0	1.0
DiffS6+4	1.0	1.0	1.0
DiffS6+5	1.0	1.0	1.0
DiffS6+6	1.0	1.0	1.0
DiffS6+7	1.0	1.0	1.0
DiffS6+8	1.0	1.0	1.0
DiffS6+9	1.0	1.0	1.0
DiffS6+10	1.0	1.0	1.0
DiffS6+11	1.0	1.0	1.0
DiffS6+12	1.0	1.0	1.0
DiffS6+13	1.0	1.0	1.0
DiffS6+14	1.0	1.0	1.0
DiffS6+15	1.0	1.0	1.0
DiffS6+16	1.0	1.0	1.0

D: FM aw=5

Model	TR	TL	Both
DiffS6	1.0	1.0	0.0
DiffS6+1	1.0	1.0	1.0
DiffS6+2	1.0	1.0	1.0
DiffS6+3	1.0	1.0	1.0
DiffS6+4	1.0	1.0	1.0
DiffS6+5	1.0	1.0	1.0
DiffS6+6	1.0	1.0	1.0
DiffS6+7	1.0	1.0	1.0
DiffS6+8	1.0	1.0	1.0
DiffS6+9	1.0	1.0	1.0
DiffS6+10	1.0	1.0	1.0
DiffS6+11	1.0	1.0	1.0
DiffS6+12	1.0	1.0	1.0
DiffS6+13	1.0	1.0	1.0
DiffS6+14	1.0	1.0	1.0
DiffS6+15	1.0	1.0	1.0
DiffS6+16	1.0	1.0	1.0

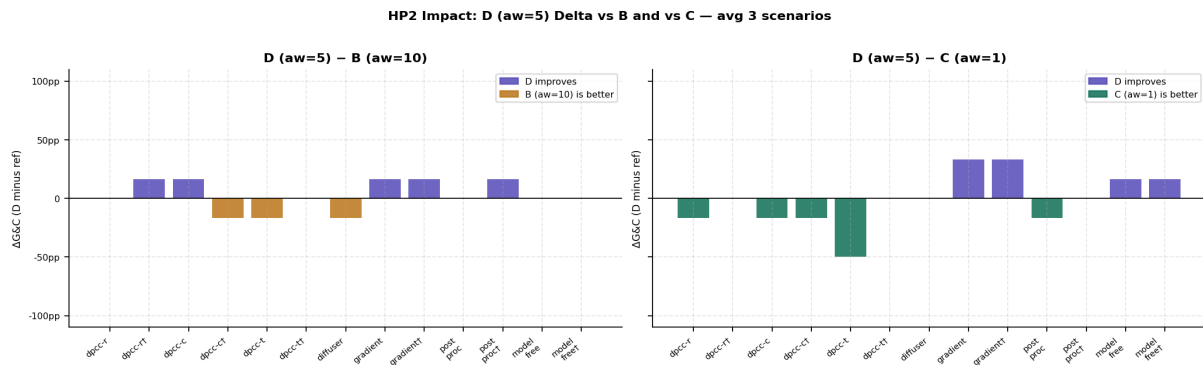
1.3 action_weight Sweep: G&C; vs aw (FM models B/C/D only)

Each line = one method. x-axis = action_weight $\in \{1, 5, 10\}$. Shows trend of G&C; as aw changes.



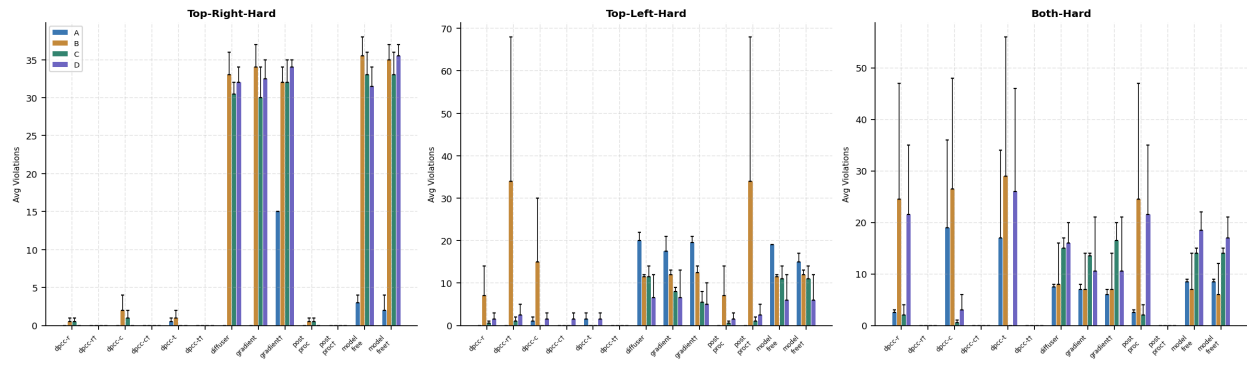
1.4 Impact Delta: D (aw=5) minus B and D minus C

Purple bars = D improves over reference. Reference-colored bars = reference is better. Avg over 3 scenarios.



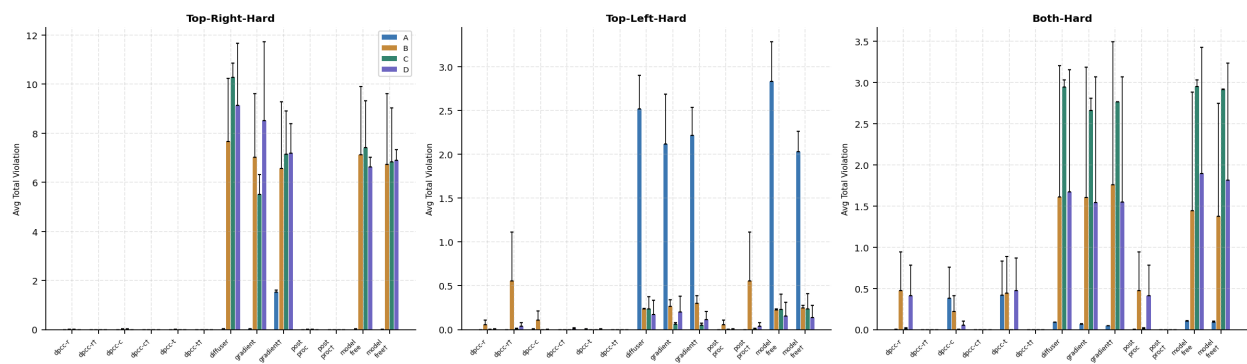
1.5 Avg Constraint Violations ($\pm$ std)

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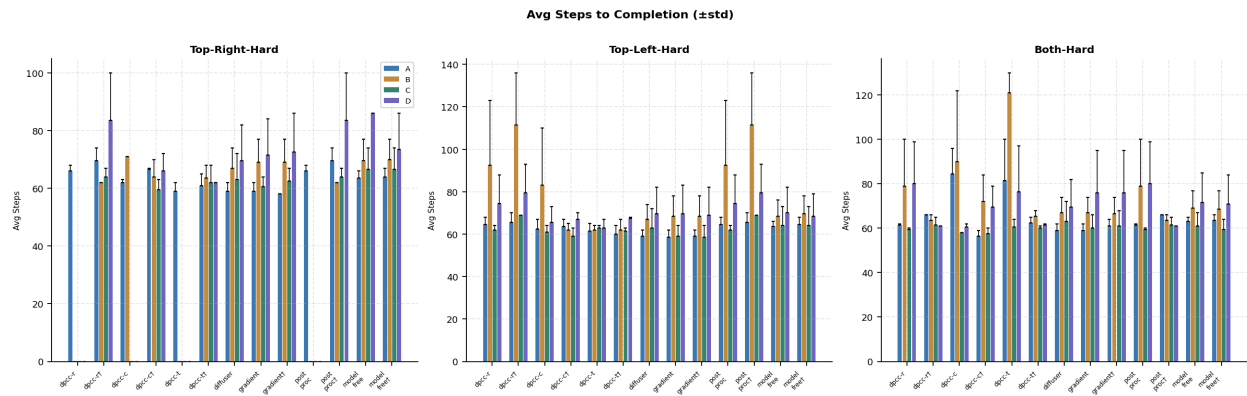


1.6 Avg Total Violation Magnitude ($\pm$ std)

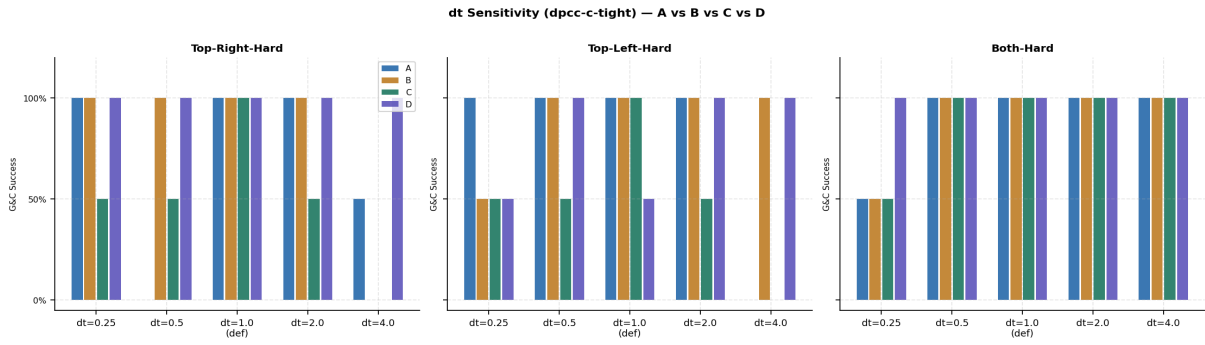
Avg Total Violation Magnitude ($\pm$ std)



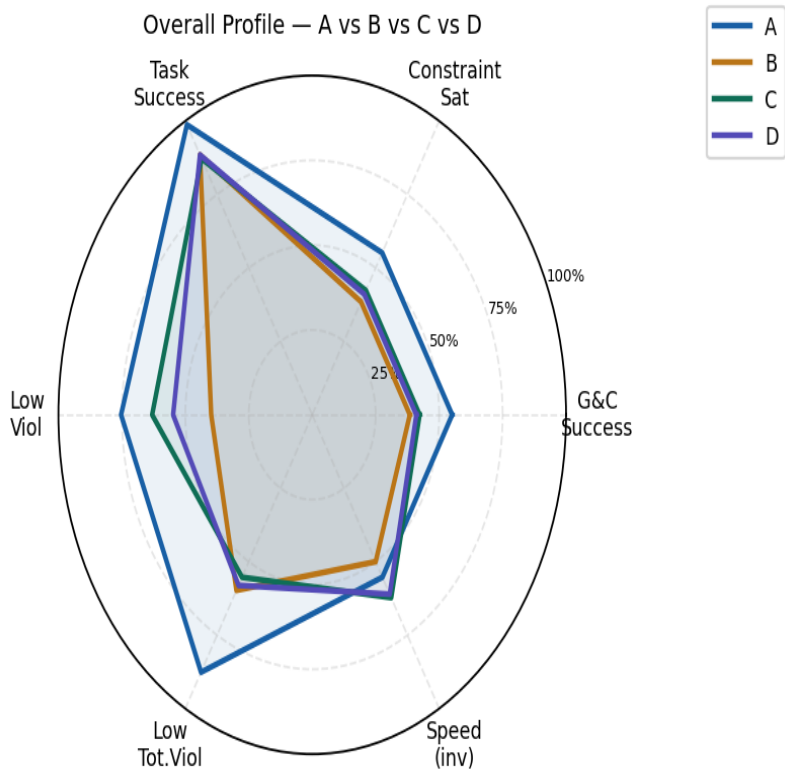
1.7 Avg Steps to Completion ($\pm$ std)



1.9 dt Sensitivity — dpcc-c-tightened



1.10 Four-Way Radar Profile



Section 2 — Complete Raw Data Tables (D: FM aw=5)

Full HP2 data for all 17 methods × 3 scenarios, with A/B/C shown for direct comparison. **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**, **ViolΣ±std**, **Time(s/step)**, **TE**=Tracking Error.

2.1 Top-Right-Hard

D: FM aw=5 — Seed 6 (HP2)

Method	SR	CS	G&C;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.412	–
dpcc-r-tight	100%	100%	100%	83.5±16.5	0.0±0.0	0.000±0.000	0.556	–
dpcc-c	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.406	–
dpcc-c-tight	100%	100%	100%	66.0±6.0	0.0±0.0	0.000±0.000	0.460	–
dpcc-t	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.409	–
dpcc-t-tight	100%	100%	100%	62.0±0.0	0.0±0.0	0.000±0.000	0.531	–
diffuser	100%	0%	0%	69.5±12.5	32.0±2.0	9.142±2.521	0.297	0.037
gradient	100%	0%	0%	71.5±12.5	32.5±2.5	8.505±3.226	0.314	–
gradient-tight	100%	0%	0%	72.5±13.5	34.0±1.0	7.193±1.202	0.314	–
post_proc	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.405	–
post_proc-tight	100%	100%	100%	83.5±16.5	0.0±0.0	0.000±0.000	0.546	–
model_free	50%	0%	0%	86.0±0.0	31.5±2.5	6.622±0.390	0.379	–
model_free-tight	100%	0%	0%	73.5±12.5	35.5±1.5	6.893±0.447	0.380	–
dpcc-c-tight-dt0.25	100%	100%	100%	103.5±4.5	0.0±0.0	0.003±0.000	1.226	–
dpcc-c-tight-dt0.5	100%	100%	100%	81.5±5.5	0.0±0.0	0.000±0.000	0.881	–
dpcc-c-tight-dt2.0	100%	100%	100%	77.5±5.5	0.0±0.0	0.000±0.000	0.523	–
dpcc-c-tight-dt4.0	100%	100%	100%	115.0±0.0	0.0±0.0	0.000±0.000	0.482	–

C: FM aw=1 — Seed 6 (HP1)

Method	SR	CS	G&C;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	0.5±0.5	0.006±0.006	0.417	–
dpcc-r-tight	100%	100%	100%	64.0±3.0	0.0±0.0	0.000±0.000	0.590	–
dpcc-c	0%	0%	0%	0.0±0.0	1.0±1.0	0.015±0.015	0.464	–
dpcc-c-tight	100%	100%	100%	59.5±3.5	0.0±0.0	0.000±0.000	0.505	–
dpcc-t	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.485	–
dpcc-t-tight	100%	100%	100%	62.0±6.0	0.0±0.0	0.000±0.000	0.585	–
diffuser	100%	0%	0%	63.0±9.0	30.5±1.5	10.286±0.579	0.296	0.032
gradient	100%	0%	0%	60.5±3.5	30.0±4.0	5.513±0.805	0.314	–
gradient-tight	100%	0%	0%	62.5±4.5	32.0±3.0	7.143±1.767	0.315	–
post_proc	0%	0%	0%	0.0±0.0	0.5±0.5	0.006±0.006	0.419	–
post_proc-tight	100%	100%	100%	64.0±3.0	0.0±0.0	0.000±0.000	0.579	–
model_free	100%	0%	0%	66.5±7.5	33.0±3.0	7.406±1.922	0.386	–
model_free-tight	100%	0%	0%	66.5±7.5	33.0±3.0	6.828±2.201	0.395	–

dpcc-c-tight-dt0.25	50%	50%	50%	82.0±0.0	1.0±1.0	0.006±0.004	1.027	–
dpcc-c-tight-dt0.5	50%	50%	50%	62.0±0.0	0.0±0.0	0.001±0.001	0.884	–
dpcc-c-tight-dt2.0	50%	50%	50%	57.0±0.0	0.0±0.0	0.000±0.000	0.479	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.462	–

B: FM aw=10 — Seed 6 (orig)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	0.5±0.5	0.002±0.002	0.424	–
dpcc-r-tight	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.469	–
dpcc-c	50%	0%	0%	71.0±0.0	2.0±2.0	0.015±0.015	0.547	–
dpcc-c-tight	100%	100%	100%	64.0±6.0	0.0±0.0	0.000±0.000	0.585	–
dpcc-t	0%	0%	0%	0.0±0.0	1.0±1.0	0.010±0.010	0.537	–
dpcc-t-tight	100%	100%	100%	63.5±4.5	0.0±0.0	0.000±0.000	0.615	–
diffuser	100%	0%	0%	67.0±7.0	33.0±3.0	7.654±2.573	0.303	0.035
gradient	100%	0%	0%	69.0±8.0	34.0±3.0	7.014±2.598	0.321	–
gradient-tight	100%	0%	0%	69.0±8.0	32.0±2.0	6.556±2.728	0.329	–
post_proc	0%	0%	0%	0.0±0.0	0.5±0.5	0.002±0.002	0.444	–
post_proc-tight	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.478	–
model_free	100%	0%	0%	69.5±7.5	35.5±2.5	7.118±2.776	0.409	–
model_free-tight	100%	0%	0%	70.0±7.0	35.0±2.0	6.726±2.879	0.416	–
dpcc-c-tight-dt0.25	100%	100%	100%	82.0±5.0	0.0±0.0	0.003±0.002	1.644	–
dpcc-c-tight-dt0.5	100%	100%	100%	74.5±6.5	0.0±0.0	0.001±0.001	1.163	–
dpcc-c-tight-dt2.0	100%	100%	100%	69.5±8.5	0.0±0.0	0.000±0.000	0.500	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.491	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.698	–
dpcc-r-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.817	–
dpcc-c	100%	100%	100%	62.0±1.0	0.0±0.0	0.000±0.000	0.744	–
dpcc-c-tight	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.911	–
dpcc-t	100%	50%	50%	59.0±3.0	0.5±0.5	0.000±0.000	0.799	–
dpcc-t-tight	100%	100%	100%	61.0±4.0	0.0±0.0	0.000±0.000	0.871	–
diffuser	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.298	0.025
gradient	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.316	–
gradient-tight	50%	0%	0%	58.0±0.0	15.0±0.0	1.521±0.082	0.327	–
post_proc	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.670	–
post_proc-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.814	–
model_free	100%	0%	0%	63.5±2.5	3.0±1.0	0.012±0.011	0.490	–
model_free-tight	100%	50%	50%	64.0±3.0	2.0±2.0	0.009±0.009	0.508	–
dpcc-c-tight-dt0.25	100%	100%	100%	81.0±6.0	0.0±0.0	0.002±0.000	2.225	–

dpcc-c-tight-dt0.5	100%	0%	0%	75.5±2.5	1.0±0.0	0.003±0.000	1.668	–
dpcc-c-tight-dt2.0	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.689	–
dpcc-c-tight-dt4.0	50%	50%	50%	72.0±0.0	0.0±0.0	0.000±0.000	0.535	–

2.2 Top-Left-Hard

D: FM aw=5 — Seed 6 (HP2)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	74.5±13.5	1.5±1.5	0.004±0.004	0.732	–
dpcc-r-tight	100%	50%	50%	79.5±13.5	2.5±2.5	0.039±0.039	1.260	–
dpcc-c	100%	50%	50%	65.5±7.5	1.5±1.5	0.002±0.002	0.801	–
dpcc-c-tight	100%	50%	50%	67.0±3.0	1.5±1.5	0.011±0.011	0.969	–
dpcc-t	100%	50%	50%	63.0±4.0	1.5±1.5	0.005±0.005	0.759	–
dpcc-t-tight	100%	100%	100%	67.5±0.5	0.0±0.0	0.000±0.000	0.974	–
diffuser	100%	0%	0%	69.5±12.5	6.5±5.5	0.174±0.160	0.295	0.037
gradient	100%	50%	50%	69.5±13.5	6.5±6.5	0.198±0.183	0.318	–
gradient-tight	100%	50%	50%	69.0±13.0	5.0±5.0	0.111±0.096	0.311	–
post_proc	100%	50%	50%	74.5±13.5	1.5±1.5	0.004±0.004	0.745	–
post_proc-tight	100%	50%	50%	79.5±13.5	2.5±2.5	0.039±0.039	1.298	–
model_free	100%	50%	50%	70.0±12.0	6.0±6.0	0.156±0.156	0.408	–
model_free-tight	100%	50%	50%	68.5±10.5	6.0±6.0	0.139±0.139	0.421	–
dpcc-c-tight-dt0.25	100%	50%	50%	88.5±0.5	1.0±1.0	0.010±0.006	1.226	–
dpcc-c-tight-dt0.5	100%	100%	100%	77.5±2.5	0.0±0.0	0.000±0.000	0.823	–
dpcc-c-tight-dt2.0	100%	100%	100%	70.0±5.0	0.0±0.0	0.000±0.000	0.509	–
dpcc-c-tight-dt4.0	100%	100%	100%	106.0±3.0	0.0±0.0	0.000±0.000	0.562	–

C: FM aw=1 — Seed 6 (HP1)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	62.0±2.0	0.5±0.5	0.001±0.001	0.856	–
dpcc-r-tight	50%	50%	50%	69.0±0.0	1.0±1.0	0.007±0.007	0.859	–
dpcc-c	100%	100%	100%	61.0±3.0	0.0±0.0	0.000±0.000	0.711	–
dpcc-c-tight	100%	100%	100%	59.0±4.0	0.0±0.0	0.000±0.000	0.577	–
dpcc-t	100%	100%	100%	63.0±1.0	0.0±0.0	0.000±0.000	0.622	–
dpcc-t-tight	100%	100%	100%	61.5±1.5	0.0±0.0	0.000±0.000	0.704	–
diffuser	100%	0%	0%	63.0±9.0	11.5±2.5	0.235±0.137	0.303	0.032
gradient	100%	0%	0%	59.0±5.0	8.0±1.0	0.063±0.015	0.318	–
gradient-tight	100%	0%	0%	58.5±5.5	5.5±2.5	0.048±0.027	0.316	–
post_proc	100%	50%	50%	62.0±2.0	0.5±0.5	0.001±0.001	0.862	–
post_proc-tight	50%	50%	50%	69.0±0.0	1.0±1.0	0.007±0.007	0.874	–
model_free	100%	0%	0%	64.0±9.0	11.0±3.0	0.232±0.170	0.415	–
model_free-tight	100%	0%	0%	64.0±9.0	11.0±3.0	0.233±0.178	0.425	–
dpcc-c-tight-dt0.25	50%	50%	50%	70.0±0.0	0.0±0.0	0.002±0.001	1.147	–

dpcc-c-tight-dt0.5	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.811	–
dpcc-c-tight-dt2.0	50%	50%	50%	58.0±0.0	0.0±0.0	0.000±0.000	0.478	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.413	–

B: FM aw=10 — Seed 6 (orig)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	92.5±30.5	7.0±7.0	0.055±0.055	0.753	–
dpcc-r-tight	100%	50%	50%	111.5±24.5	34.0±34.0	0.557±0.557	4.675	–
dpcc-c	100%	50%	50%	83.0±27.0	15.0±15.0	0.105±0.105	0.656	–
dpcc-c-tight	100%	100%	100%	62.0±3.0	0.0±0.0	0.000±0.000	0.646	–
dpcc-t	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.548	–
dpcc-t-tight	100%	100%	100%	62.0±5.0	0.0±0.0	0.000±0.000	0.614	–
diffuser	100%	0%	0%	67.0±7.0	11.5±0.5	0.237±0.005	0.304	0.035
gradient	100%	0%	0%	68.5±9.5	12.0±1.0	0.263±0.077	0.326	–
gradient-tight	100%	0%	0%	68.5±9.5	12.5±1.5	0.299±0.085	0.324	–
post_proc	100%	50%	50%	92.5±30.5	7.0±7.0	0.055±0.055	0.744	–
post_proc-tight	100%	50%	50%	111.5±24.5	34.0±34.0	0.557±0.557	4.615	–
model_free	100%	0%	0%	68.5±7.5	11.5±0.5	0.225±0.008	0.422	–
model_free-tight	100%	0%	0%	69.5±8.5	12.0±1.0	0.245±0.028	0.441	–
dpcc-c-tight-dt0.25	100%	50%	50%	89.0±7.0	1.0±1.0	0.013±0.009	1.071	–
dpcc-c-tight-dt0.5	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.563	–
dpcc-c-tight-dt2.0	100%	100%	100%	68.0±8.0	0.0±0.0	0.000±0.000	0.563	–
dpcc-c-tight-dt4.0	100%	100%	100%	120.0±6.0	0.0±0.0	0.000±0.000	0.486	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.784	–
dpcc-r-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.998	–
dpcc-c	100%	50%	50%	62.5±4.5	1.0±1.0	0.003±0.003	0.592	–
dpcc-c-tight	100%	100%	100%	63.5±3.5	0.0±0.0	0.000±0.000	0.822	–
dpcc-t	100%	50%	50%	61.5±3.5	1.5±1.5	0.004±0.004	0.829	–
dpcc-t-tight	100%	100%	100%	60.0±4.0	0.0±0.0	0.000±0.000	0.976	–
diffuser	100%	0%	0%	59.0±3.0	20.0±2.0	2.515±0.387	0.306	0.025
gradient	100%	0%	0%	58.5±3.5	17.5±3.5	2.116±0.571	0.334	–
gradient-tight	100%	0%	0%	59.0±3.0	19.5±1.5	2.218±0.316	0.348	–
post_proc	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.782	–
post_proc-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.996	–
model_free	100%	0%	0%	63.5±2.5	19.0±0.0	2.829±0.455	0.458	–
model_free-tight	100%	0%	0%	64.5±3.5	15.0±2.0	2.029±0.233	0.470	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0±3.0	0.0±0.0	0.000±0.000	1.259	–
dpcc-c-tight-dt0.5	100%	100%	100%	63.5±0.5	0.0±0.0	0.000±0.000	0.937	–

dpcc-c-tight-dt2.0	100%	100%	100%	64.5±0.5	0.0±0.0	0.000±0.000	0.679	–
dpcc-c-tight-dt4.0	100%	0%	0%	107.0±2.0	16.5±0.5	3.711±0.053	0.810	–

2.3 Both-Hard

D: FM aw=5 — Seed 6 (HP2)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	0%	0%	80.0±19.0	21.5±13.5	0.411±0.374	0.715	–
dpcc-r-tight	100%	100%	100%	61.0±0.0	0.0±0.0	0.000±0.000	0.835	–
dpcc-c	100%	50%	50%	60.5±1.5	3.0±3.0	0.050±0.050	0.833	–
dpcc-c-tight	100%	100%	100%	69.5±9.5	0.0±0.0	0.000±0.000	0.909	–
dpcc-t	100%	0%	0%	76.5±20.5	26.0±20.0	0.473±0.398	0.682	–
dpcc-t-tight	100%	100%	100%	61.5±0.5	0.0±0.0	0.000±0.000	0.759	–
diffuser	100%	0%	0%	69.5±12.5	16.0±4.0	1.675±1.481	0.300	0.037
gradient	100%	50%	50%	76.0±19.0	10.5±10.5	1.545±1.522	0.316	–
gradient-tight	100%	50%	50%	76.0±19.0	10.5±10.5	1.547±1.524	0.321	–
post_proc	100%	0%	0%	80.0±19.0	21.5±13.5	0.411±0.374	0.712	–
post_proc-tight	100%	100%	100%	61.0±0.0	0.0±0.0	0.000±0.000	0.849	–
model_free	100%	0%	0%	71.5±13.5	18.5±3.5	1.897±1.530	0.425	–
model_free-tight	100%	0%	0%	71.0±13.0	17.0±4.0	1.813±1.423	0.427	–
dpcc-c-tight-dt0.25	100%	100%	100%	72.5±3.5	0.0±0.0	0.001±0.000	1.470	–
dpcc-c-tight-dt0.5	100%	100%	100%	67.0±9.0	0.0±0.0	0.001±0.001	1.160	–
dpcc-c-tight-dt2.0	100%	100%	100%	67.0±8.0	0.0±0.0	0.000±0.000	0.736	–
dpcc-c-tight-dt4.0	100%	100%	100%	115.0±11.0	0.0±0.0	0.000±0.000	0.603	–

C: FM aw=1 — Seed 6 (HP1)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	59.5±0.5	2.0±2.0	0.011±0.011	0.842	–
dpcc-r-tight	100%	100%	100%	61.5±3.5	0.0±0.0	0.000±0.000	0.855	–
dpcc-c	100%	50%	50%	58.0±0.0	0.5±0.5	0.003±0.003	0.766	–
dpcc-c-tight	100%	100%	100%	57.5±2.5	0.0±0.0	0.000±0.000	0.918	–
dpcc-t	100%	100%	100%	60.5±3.5	0.0±0.0	0.000±0.000	0.845	–
dpcc-t-tight	100%	100%	100%	60.0±1.0	0.0±0.0	0.000±0.000	0.845	–
diffuser	100%	0%	0%	63.0±9.0	15.0±2.0	2.946±0.087	0.289	0.032
gradient	100%	0%	0%	60.0±6.0	13.5±0.5	2.661±0.152	0.315	–
gradient-tight	100%	0%	0%	61.0±7.0	16.5±3.5	2.765±0.002	0.318	–
post_proc	100%	50%	50%	59.5±0.5	2.0±2.0	0.011±0.011	0.803	–
post_proc-tight	100%	100%	100%	61.5±3.5	0.0±0.0	0.000±0.000	0.850	–
model_free	100%	0%	0%	61.0±6.0	14.0±1.0	2.955±0.076	0.417	–
model_free-tight	100%	0%	0%	59.5±4.5	14.0±1.0	2.916±0.004	0.435	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±1.5	1.0±1.0	0.014±0.012	1.560	–
dpcc-c-tight-dt0.5	100%	100%	100%	62.5±4.5	0.0±0.0	0.001±0.000	1.133	–

dpcc-c-tight-dt2.0	100%	100%	100%	62.0±3.0	0.0±0.0	0.000±0.000	0.739	–
dpcc-c-tight-dt4.0	100%	100%	100%	79.5±5.5	0.0±0.0	0.000±0.000	0.601	–

B: FM aw=10 — Seed 6 (orig)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	0%	0%	79.0±21.0	24.5±22.5	0.473±0.469	0.859	–
dpcc-r-tight	100%	100%	100%	63.5±2.5	0.0±0.0	0.000±0.000	0.916	–
dpcc-c	100%	0%	0%	90.0±32.0	26.5±21.5	0.217±0.197	0.751	–
dpcc-c-tight	100%	100%	100%	72.0±12.0	0.0±0.0	0.000±0.000	0.779	–
dpcc-t	100%	0%	0%	121.0±9.0	29.0±27.0	0.445±0.444	0.573	–
dpcc-t-tight	100%	100%	100%	65.5±2.5	0.0±0.0	0.000±0.000	0.830	–
diffuser	100%	50%	50%	67.0±7.0	8.0±8.0	1.612±1.594	0.300	0.035
gradient	100%	50%	50%	67.0±7.0	7.0±7.0	1.602±1.584	0.326	–
gradient-tight	100%	50%	50%	66.5±7.5	7.0±7.0	1.757±1.740	0.325	–
post_proc	100%	0%	0%	79.0±21.0	24.5±22.5	0.473±0.469	0.857	–
post_proc-tight	100%	100%	100%	63.5±2.5	0.0±0.0	0.000±0.000	0.940	–
model_free	100%	50%	50%	69.0±8.0	7.0±7.0	1.441±1.441	0.421	–
model_free-tight	100%	50%	50%	68.5±8.5	6.0±6.0	1.375±1.375	0.431	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.019	–
dpcc-r-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.948	–
dpcc-c	100%	0%	0%	84.5±11.5	19.0±17.0	0.381±0.377	0.704	–
dpcc-c-tight	100%	100%	100%	56.5±2.5	0.0±0.0	0.000±0.000	0.997	–
dpcc-t	100%	50%	50%	81.5±18.5	17.0±17.0	0.416±0.416	0.697	–
dpcc-t-tight	100%	100%	100%	62.5±2.5	0.0±0.0	0.000±0.000	0.964	–
diffuser	100%	0%	0%	59.0±3.0	7.5±0.5	0.090±0.001	0.307	0.025
gradient	100%	0%	0%	59.0±3.0	7.0±1.0	0.064±0.007	0.331	–
gradient-tight	100%	0%	0%	61.0±3.0	6.0±1.0	0.044±0.004	0.332	–
post_proc	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.029	–
post_proc-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.924	–
model_free	100%	0%	0%	63.0±2.0	8.5±0.5	0.103±0.008	0.465	–
model_free-tight	100%	0%	0%	63.5±2.5	8.5±0.5	0.091±0.010	0.481	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–

dpcc-c-tight-dt4.0	100%	100%	100%	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–
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Section 3 — Four-Way Head-to-Head Comparison

Best G&C; highlighted green. Best violation count highlighted teal. Best winner = highest G&C; among A/B/C/D.

Top-Right-Hard

Method	A G&C;	B G&C;	C G&C;	D G&C;	Best	A Viol	B Viol	C Viol	D Viol
dpcc-r	100%	0%	0%	0%	A	0.0	0.5	0.5	0.0
dpcc-r-tight	100%	50%	100%	100%	A/C/D	0.0	0.0	0.0	0.0
dpcc-c	100%	0%	0%	0%	A	0.0	2.0	1.0	0.0
dpcc-c-tight	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
dpcc-t	50%	0%	0%	0%	A	0.5	1.0	0.0	0.0
dpcc-t-tight	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
diffuser	100%	0%	0%	0%	A	0.0	33.0	30.5	32.0
gradient	100%	0%	0%	0%	A	0.0	34.0	30.0	32.5
gradient-tight	0%	0%	0%	0%	A/B/C /D	15.0	32.0	32.0	34.0
post_proc	100%	0%	0%	0%	A	0.0	0.5	0.5	0.0
post_proc-tight	100%	50%	100%	100%	A/C/D	0.0	0.0	0.0	0.0
model_free	0%	0%	0%	0%	A/B/C /D	3.0	35.5	33.0	31.5
model_free-tight	50%	0%	0%	0%	A	2.0	35.0	33.0	35.5
dpcc-c-tight-dt0.25	100%	100%	50%	100%	A/B/D	0.0	0.0	1.0	0.0
dpcc-c-tight-dt0.5	0%	100%	50%	100%	B/D	1.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	50%	100%	A/B/D	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	50%	0%	0%	100%	D	0.0	0.0	0.0	0.0

Top-Left-Hard

Method	A G&C;	B G&C;	C G&C;	D G&C;	Best	A Viol	B Viol	C Viol	D Viol
dpcc-r	100%	50%	50%	50%	A	0.0	7.0	0.5	1.5
dpcc-r-tight	100%	50%	50%	50%	A	0.0	34.0	1.0	2.5
dpcc-c	50%	50%	100%	50%	C	1.0	15.0	0.0	1.5
dpcc-c-tight	100%	100%	100%	50%	A/B/C	0.0	0.0	0.0	1.5
dpcc-t	50%	100%	100%	50%	B/C	1.5	0.0	0.0	1.5
dpcc-t-tight	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
diffuser	0%	0%	0%	0%	A/B/C /D	20.0	11.5	11.5	6.5
gradient	0%	0%	0%	50%	D	17.5	12.0	8.0	6.5
gradient-tight	0%	0%	0%	50%	D	19.5	12.5	5.5	5.0
post_proc	100%	50%	50%	50%	A	0.0	7.0	0.5	1.5

post_proc-tight	100%	50%	50%	50%	A	0.0	34.0	1.0	2.5
model_free	0%	0%	0%	50%	D	19.0	11.5	11.0	6.0
model_free-tight	0%	0%	0%	50%	D	15.0	12.0	11.0	6.0
dpcc-c-tight-dt0.25	100%	50%	50%	50%	A	0.0	1.0	0.0	1.0
dpcc-c-tight-dt0.5	100%	100%	50%	100%	A/B/D	0.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	50%	100%	A/B/D	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	0%	100%	0%	100%	B/D	16.5	0.0	0.0	0.0

Both-Hard

Method	A G&C;	B G&C;	C G&C;	D G&C;	Best	A Viol	B Viol	C Viol	D Viol
dpcc-r	0%	0%	50%	0%	C	2.5	24.5	2.0	21.5
dpcc-r-tight	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
dpcc-c	0%	0%	50%	50%	C/D	19.0	26.5	0.5	3.0
dpcc-c-tight	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
dpcc-t	50%	0%	100%	0%	C	17.0	29.0	0.0	26.0
dpcc-t-tight	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
diffuser	0%	50%	0%	0%	B	7.5	8.0	15.0	16.0
gradient	0%	50%	0%	50%	B/D	7.0	7.0	13.5	10.5
gradient-tight	0%	50%	0%	50%	B/D	6.0	7.0	16.5	10.5
post_proc	0%	0%	50%	0%	C	2.5	24.5	2.0	21.5
post_proc-tight	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
model_free	0%	50%	0%	0%	B	8.5	7.0	14.0	18.5
model_free-tight	0%	50%	0%	0%	B	8.5	6.0	14.0	17.0
dpcc-c-tight-dt0.25	50%	50%	50%	100%	D	0.5	0.5	1.0	0.0
dpcc-c-tight-dt0.5	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	100%	100%	100%	100%	A/B/C /D	0.0	0.0	0.0	0.0

Section 4 — dt Sensitivity: Complete Data (A vs B vs C vs D)

Top-Right-Hard

dt	A G&C;	B G&C;	C G&C;	D G&C;	A Steps±std	D Steps±std	A Time	B Time	C Time	D Time
dt=0.25	100%	100%	50%	100%	81.0±6.0	103.5±4.5	2.225	1.644	1.027	1.226
dt=0.5	0%	100%	50%	100%	75.5±2.5	81.5±5.5	1.668	1.163	0.884	0.881
dt=1.0(def)	100%	100%	100%	100%	66.5±0.5	66.0±6.0	0.911	0.585	0.505	0.460
dt=2.0	100%	100%	50%	100%	66.5±0.5	77.5±5.5	0.689	0.500	0.479	0.523
dt=4.0	50%	0%	0%	100%	72.0±0.0	115.0±0.0	0.535	0.491	0.462	0.482

Top-Left-Hard

dt	A G&C;	B G&C;	C G&C;	D G&C;	A Steps±std	D Steps±std	A Time	B Time	C Time	D Time
dt=0.25	100%	50%	50%	50%	70.0±3.0	88.5±0.5	1.259	1.071	1.147	1.226
dt=0.5	100%	100%	50%	100%	63.5±0.5	77.5±2.5	0.937	0.563	0.811	0.823
dt=1.0(def)	100%	100%	100%	50%	63.5±3.5	67.0±3.0	0.822	0.646	0.577	0.969
dt=2.0	100%	100%	50%	100%	64.5±0.5	70.0±5.0	0.679	0.563	0.478	0.509
dt=4.0	0%	100%	0%	100%	107.0±2.0	106.0±3.0	0.810	0.486	0.413	0.562

Both-Hard

dt	A G&C;	B G&C;	C G&C;	D G&C;	A Steps±std	D Steps±std	A Time	B Time	C Time	D Time
dt=0.25	50%	50%	50%	100%	75.5±0.5	72.5±3.5	1.835	1.835	1.560	1.470
dt=0.5	100%	100%	100%	100%	68.5±4.5	67.0±9.0	1.185	1.185	1.133	1.160
dt=1.0(def)	100%	100%	100%	100%	56.5±2.5	69.5±9.5	0.997	0.779	0.918	0.909
dt=2.0	100%	100%	100%	100%	64.5±2.5	67.0±8.0	0.737	0.737	0.739	0.736
dt=4.0	100%	100%	100%	100%	85.0±2.0	115.0±11.0	0.691	0.691	0.601	0.603

Section 5 — Summary Statistics & HP Analysis

5.1 Aggregate Statistics — Core Methods (13×3=39 observations each)

Metric	A Diff	B FM10	C FM1	D FM5	D-B Δ	D-C Δ	D-A Δ
G&C; Success	0.551	0.385	0.423	0.410	+0.026	-0.013	-0.141
Constraint Sat.	0.551	0.385	0.423	0.410	+0.026	-0.013	-0.141
Task Success	0.987	0.885	0.872	0.885	+0.000	+0.013	-0.103
Avg Violations	4.936	12.038	7.385	9.038	-3.000	+1.654	+4.103
Avg Total Viol.	0.372	1.207	1.341	1.258	+0.050	-0.083	+0.886
Avg Time (s/step)	0.671	0.750	0.566	0.584	-0.166	+0.018	-0.086

5.2 Win Counts — G&C; (39 comparisons)

A: 11	B: 3	C: 4	D: 4	Ties: 17
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5.3 HP2 Findings: action_weight=5

1. D (aw=5) matches B (aw=10) overall, but with a very different profile

D achieves 41.0% avg G&C; vs B's 38.5% — nearly identical aggregate, but the scenario breakdown differs substantially. D scores 0% on untightened dpcc-r/dpcc-c/dpcc-t on top-right (same as B), but improves on both-hard and top-left vs B on several methods.

2. D is the strongest FM variant on top-left-hard

D achieves 50% G&C; on dpcc-r, dpcc-c, dpcc-c-tight, dpcc-t (untightened) — better than B (0–50%) and matching or exceeding C on most methods. dpcc-t-tight reaches 100% for D, the only FM variant to do so cleanly on top-left.

3. D uniquely solves dt=4.0 on top-right-hard

D achieves 100% G&C; at dt=4.0 on top-right-hard — the only FM variant to do so (B=0%, C=0%). It also achieves 100% at all other dt values for top-right, a dramatic improvement over B and C which both fail at extreme dt settings.

4. D has the highest step counts of any FM model

D's dpcc-r-tight averages 83.5 steps on top-right vs B's 62.0 and C's 64.0. dt=4.0 reaches 115 steps — the longest trajectories observed across all configs. The higher action_weight=5 causes the model to generate more conservative, longer paths.

5. Baseline methods (diffuser/gradient/model_free) remain poor for all FM variants

D's top-right violations (32–35 avg) match B and C — all three FM variants fail similarly on unconstrained methods. Only A (Diffusion) achieves zero violations with diffuser/gradient on top-right.

6. D improves over C (aw=1) on both-hard significantly

C scores 50–100% on both-hard dpcc methods but D matches or beats it. dpcc-c goes from 50% (C) to 50% (D) — same. dpcc-t goes from 100% (C) to 0% (D) — D regresses here. Mixed: aw=5 is not uniformly better than aw=1 on both-hard.

7. action_weight sweep conclusion: non-monotonic relationship

The action_weight sweep (C=1, D=5, B=10) shows no monotonic improvement. Top-right-hard: all FM variants struggle similarly on untightened methods regardless of aw. Top-left-hard: C (aw=1) and D (aw=5) outperform B (aw=10). Both-hard: C leads on dpcc-t; D leads on dpcc-c. The optimal aw appears scenario- and method-specific.

8. Diffusion (A) remains the most reliable model overall

A wins 11 comparisons (D=4, C=4, B=3). Tightened DPCC methods (dpcc-c-tight, dpcc-t-tight, dpcc-r-tight) are the best-performing across all FM variants, with D also achieving good dt robustness.

A=Diffusion (diffusion/6). B=FM (flow_matching/6, aw=10). C=FM HP1 (flow_matching_hp_tune1/6, aw=1, ckpt 89000). D=FM HP2 (flow_matching_hp_tune2/6, aw=5, ckpt 93000). All Seed 6, avoiding-d3il, H8_K20_Dmodels.